

		WIKITEXT-103			PUBMED			WIKI40B-Arabic		
		Original	Mean Embed.	Ours	Original	Mean Embed.	Ours	Original	Mean Embed.	Ours
New words	new token	–	0.071	0.171	–	0.123	0.180	–	0.117	0.402
	original or new token	0.322	0.178	0.284	0.280	0.171	0.259	0.413	0.119	0.414
All words		0.522	0.473	0.519	0.517	0.479	0.511	0.535	0.211	0.532

Table 1: Token-level accuracy of finetuning-free vocabulary expansion for Llama2-7B on three datasets. We compare our method using detokenized representations (*Ours*) to the original model (*Original*) and to an expansion baseline using the word’s average token embeddings from *E* and *U* (*Mean Embed.*; Gee et al. 2022). Accuracy is reported for newly added words, where we distinguish between correctly predicting the new tokens (top row) and predicting either the word’s original first token, as the unexpanded model would, or its new token (middle row). We also report overall performance on all tokens (last row). Our method enables the frozen model to use the newly added input and output embeddings, while maintaining overall model performance.

We apply our approach to LLAMA2-7B and experiment with three datasets: WIKITEXT-103 (Merity et al., 2017), abstracts of biomedical articles from PUBMED (Xiong et al., 2024), and the Arabic split of WIKI40B (Guo et al., 2020). For each dataset, we expand the model’s vocabulary with all multi-token words that appear at least m times in the test set.¹⁷ We then learn the refinement matrices W_E and W_U using 20M tokens from the train set.¹⁸ We evaluate models in a next-word prediction setup and compare our approach against two baselines: the original model with an unmodified vocabulary (*original*); and an expanded model that follows our framework but uses the mean embeddings of the word’s tokens in *E* and *U* to initialize its new representation, instead of the detokenized representations r , following Gee et al. (2022; *mean embedding*). For evaluation, perplexity is not a suitable metric, as differences in vocabulary between the approaches skew perplexity scores and prevent a fair comparison. Instead, we measure the model’s token-level top-1 accuracy when it is given each token’s previous context.

Our results (Table 1) indicate that models can generalize to new vocabulary entries surprisingly well when these are initialized with their own detokenized representations. Unlike the *mean embeddings* expansion baseline, which both struggles with using new tokens and degrades overall performance, our method allows the model to successfully integrate new vocabulary words while preserving its accuracy on existing words. This effect is particularly pronounced in Arabic WIKI40B, where the model almost always selects new tokens instead of the original tokenization (40.2% vs 41.3%), demonstrating the potential of our approach for multilingual and domain-specific applications.

Our method also shows potential to reduce the number of tokens processed during encoding and inference: across the three datasets, we observe a reduction in the total number of tokens processed during encoding by 10.5% to 14.5% (see Table 5 in App. H), while maintaining model performance. In future work, we will further explore the application of our framework to multilingual adaptation (Petrov et al., 2023; Alabi et al., 2022) and continual domain-adaptive pretraining (Ke et al., 2023; Yıldız et al., 2024; Gururangan et al., 2020).

7 CONCLUSION

The ability of LLMs to comprehend and generate language relies on intricate internal processes, and understanding these mechanisms is crucial for improving model performance and efficiency. In this work, we unraveled the word detokenization process, shedding light on how models internally transform fragmented sub-word tokens into coherent word representations formed at the last token. Our results indicate that this mechanism manifests in early to middle layers, where models attempt to reconstruct words by mapping them to an inner lexicon using their FFN layers. We provided evidence this lexicon is more exhaustive than the tokenizer’s vocabulary, and could help models to recognize words even amidst noise.

Our work also unlocks practical avenues for optimizing tokenization, as well as the speed and cost of inference. We demonstrated one such application and presented a finetuning-free method to expand the vocabulary of LLMs. We hope our work paves the way for more efficient and versatile models.

¹⁷We use $m = 1, 5, 50$ for WIKITEXT-103, PUBMED and Arabic WIKI40B respectively.

¹⁸We use a sequence length of 512 and train on 10,000 sequences, taking up to 30 minutes on a single GPU.

ACKNOWLEDGMENTS

This work was supported in part by the Israel Science Foundation (grant no. 2045/21) and by NSF-BSF grant 2020793. We are grateful to Iddo Yosha for his valuable insights and thank the anonymous reviewers for their constructive feedback.

REFERENCES

- Orevaoghene Ahia, Sachin Kumar, Hila Gonen, Jungo Kasai, David Mortensen, Noah Smith, and Yulia Tsvetkov. Do all languages cost the same? tokenization in the era of commercial language models. In Houda Bouamor, Juan Pino, and Kalika Bali (eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 9904–9923, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.614. URL <https://aclanthology.org/2023.emnlp-main.614>.
01. AI, :, Alex Young, Bei Chen, Chao Li, Chengen Huang, Ge Zhang, Guanwei Zhang, Heng Li, Jiangcheng Zhu, Jianqun Chen, Jing Chang, Kaidong Yu, Peng Liu, Qiang Liu, Shawn Yue, Senbin Yang, Shiming Yang, Tao Yu, Wen Xie, Wenhao Huang, Xiaohui Hu, Xiaoyi Ren, Xinyao Niu, Pengcheng Nie, Yuchi Xu, Yudong Liu, Yue Wang, Yuxuan Cai, Zhenyu Gu, Zhiyuan Liu, and Zonghong Dai. Yi: Open foundation models by 01.ai, 2024. URL <https://arxiv.org/abs/2403.04652>.
- Jean Aitchison. *Words in the mind: An introduction to the mental lexicon*. John Wiley & Sons, 2012.
- Jesujoba O. Alabi, David Ifeoluwa Adelani, Marius Mosbach, and Dietrich Klakow. Adapting pre-trained language models to African languages via multilingual adaptive fine-tuning. In Nicoletta Calzolari, Chu-Ren Huang, Hansaem Kim, James Pustejovsky, Leo Wanner, Key-Sun Choi, Pum-Mo Ryu, Hsin-Hsi Chen, Lucia Donatelli, Heng Ji, Sadao Kurohashi, Patrizia Paggio, Nianwen Xue, Seokhwan Kim, Younggyun Hahm, Zhong He, Tony Kyungil Lee, Enrico Santus, Francis Bond, and Seung-Hoon Na (eds.), *Proceedings of the 29th International Conference on Computational Linguistics*, pp. 4336–4349, Gyeongju, Republic of Korea, October 2022. International Committee on Computational Linguistics. URL <https://aclanthology.org/2022.coling-1.382/>.
- Khuyagbaatar Batsuren, Ekaterina Vylomova, Verna Dankers, Tsetsuukhei Delgerbaatar, Omri Uzan, Yuval Pinter, and Gábor Bella. Evaluating subword tokenization: Alien subword composition and oov generalization challenge. *arXiv preprint arXiv:2404.13292*, 2024.
- Thomas Bauwens and Pieter Delobelle. Bpe-knockout: Pruning pre-existing bpe tokenisers with backwards-compatible morphological semi-supervision. *Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2024.
- Nora Belrose, Zach Furman, Logan Smith, Danny Halawi, Igor Ostrovsky, Lev McKinney, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens, 2023. URL <https://arxiv.org/abs/2303.08112>.
- Amit Ben Artzy and Roy Schwartz. Attend first, consolidate later: On the importance of attention in different LLM layers. In Yonatan Belinkov, Najoung Kim, Jaap Jumelet, Hosein Mohebbi, Aaron Mueller, and Hanjie Chen (eds.), *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pp. 177–184, Miami, Florida, US, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.blackboxnlp-1.10. URL <https://aclanthology.org/2024.blackboxnlp-1.10/>.
- Kaj Bostrom and Greg Durrett. Byte pair encoding is suboptimal for language model pretraining. In Trevor Cohn, Yulan He, and Yang Liu (eds.), *Findings of the Association for Computational Linguistics: EMNLP 2020*, pp. 4617–4624, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.findings-emnlp.414. URL <https://aclanthology.org/2020.findings-emnlp.414>.
- Qi Cao, Takeshi Kojima, Yutaka Matsuo, and Yusuke Iwasawa. Unnatural error correction: GPT-4 can almost perfectly handle unnatural scrambled text. In Houda Bouamor, Juan Pino, and Kalika Bali (eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language*

- Processing*, pp. 8898–8913, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.550. URL <https://aclanthology.org/2023.emnlp-main.550>.
- Kenneth Ward Church. Emerging trends: Subwords, seriously? *Natural Language Engineering*, 26(3):375–382, 2020.
- Alexis Conneau, German Kruszewski, Guillaume Lample, Loïc Barrault, and Marco Baroni. What you can cram into a single $\$ \&! \#^*$ vector: Probing sentence embeddings for linguistic properties. In Iryna Gurevych and Yusuke Miyao (eds.), *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 2126–2136, Melbourne, Australia, July 2018. Association for Computational Linguistics. doi: 10.18653/v1/P18-1198. URL <https://aclanthology.org/P18-1198>.
- Gonalo M. Correia, Vlad Niculae, and Andr  F. T. Martins. Adaptively sparse transformers. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan (eds.), *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pp. 2174–2184, Hong Kong, China, November 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1223. URL <https://aclanthology.org/D19-1223/>.
- Damai Dai, Li Dong, Yaru Hao, Zhifang Sui, Baobao Chang, and Furu Wei. Knowledge neurons in pretrained transformers. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (eds.), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 8493–8502, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.581. URL <https://aclanthology.org/2022.acl-long.581>.
- Guy Dar, Mor Geva, Ankit Gupta, and Jonathan Berant. Analyzing transformers in embedding space. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki (eds.), *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 16124–16170, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.893. URL <https://aclanthology.org/2023.acl-long.893>.
- Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Nadir Durrani, Hassan Sajjad, Fahim Dalvi, and Yonatan Belinkov. Analyzing individual neurons in pre-trained language models. In Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu (eds.), *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 4865–4880, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-main.395. URL <https://aclanthology.org/2020.emnlp-main.395>.
- Nelson Elhage, Tristan Hume, Catherine Olsson, Neel Nanda, Tom Henighan, Scott Johnston, Sheer ElShowk, Nicholas Joseph, Nova DasSarma, Ben Mann, Danny Hernandez, Amanda Askell, Kamal Ndousse, Andy Jones, Dawn Drain, Anna Chen, Yuntao Bai, Deep Ganguli, Liane Lovitt, Zac Hatfield-Dodds, Jackson Kernion, Tom Conerly, Shauna Kravec, Stanislaw Fort, Saurav Kadavath, Josh Jacobson, Eli Tran-Johnson, Jared Kaplan, Jack Clark, Tom Brown, Sam McCandlish, Dario Amodei, and Christopher Olah. Softmax linear units. *Transformer Circuits Thread*, 2022. <https://transformer-circuits.pub/2022/solu/index.html>.
- Javier Ferrando and Elena Voita. Information flow routes: Automatically interpreting language models at scale. *arXiv preprint arXiv:2403.00824*, 2024.
- Sheridan Feucht, David Atkinson, Byron C Wallace, and David Bau. Token erasure as a footprint of implicit vocabulary items in LLMs. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 9727–9739, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.543. URL <https://aclanthology.org/2024.emnlp-main.543/>.